

Trainings ▾

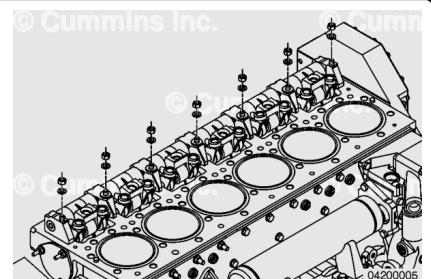
Preparatory Steps

- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/35/35-002-004-tr.html>)



Remove

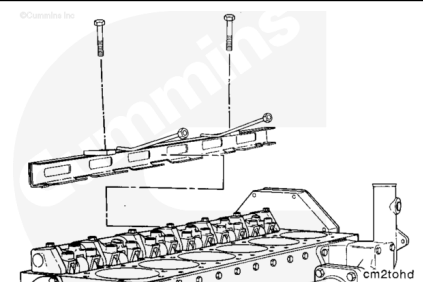
Remove the nuts and washers from the cam follower assembly supports.



Install the cam follower assembly installation and removal tool, Part Number 3824519. Slide the tool under the assemblies with the supports located in the cutouts on the bottom surface of the tool.

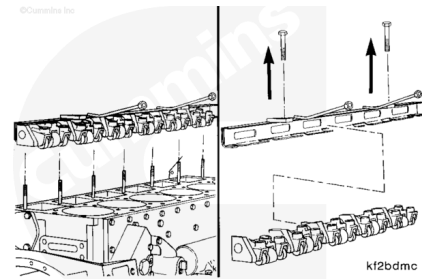
The cam follower assembly consists of two shaft assemblies with a common center support.

Secure the tool to the assemblies by hand-tightening the screws on top of the tool.



Lift both ends of the cam follower assemblies to remove them from the engine.

Set the assemblies on a flat surface. Loosen the two screws on top of the tool, and remove the tool from the cam follower assemblies.

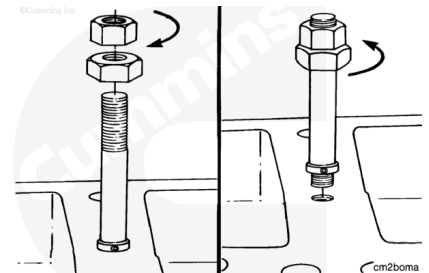


Install two M10 x 1.50 nuts on each stud.

Tighten the nuts together.

Turn the bottom nut **counterclockwise** to remove the stud.

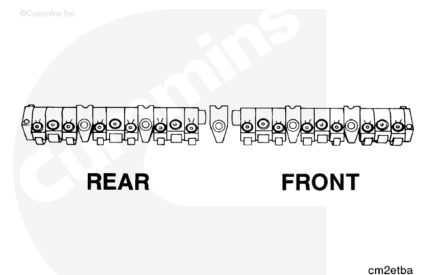
Note : The cam follower studs have an internal oil passage to supply the cam followers with lubricating oil. The studs **must** be removed and checked.



Disassemble

Separate the two shafts from the center support.

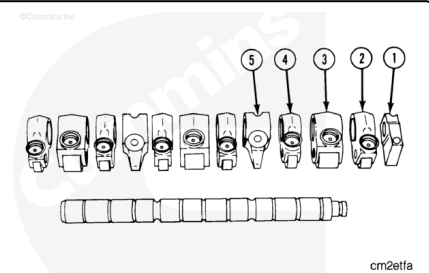
The cam follower levers **must** be installed in the same position they were removed from. Mark both of the end supports, the center support, and all of the cam followers to identify their location when they are removed. The end supports are **not** interchangeable. The center support is **not** interchangeable with any of the other inner supports.



Remove the end supports (1) from the shafts.

Remove the cam followers and supports from both shafts.

- Valve cam follower (2)
- Injector cam follower (3)
- Valve cam follower (4)
- Shaft support (5).



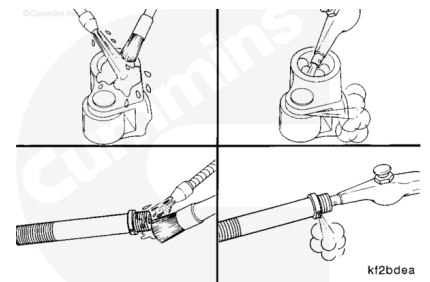
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

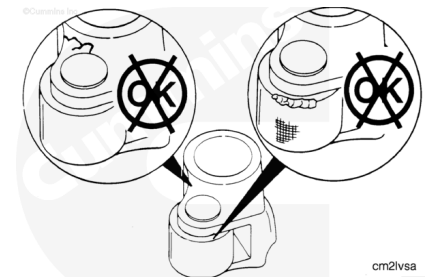


Clean the cam follower parts with solvent and dry with compressed air.

Make sure the oil passages in the cam followers and the cam follower studs are clean.

Inspect the cam followers for cracks or other damage.

Refer to Camshaft Reuse Guidelines, Bulletin 3666052 ([/qs3/pubsys2/xml/en/bulletin/3666052.html](http://qs3/pubsys2/xml/en/bulletin/3666052.html)), for inspection criteria of cam follower rollers.



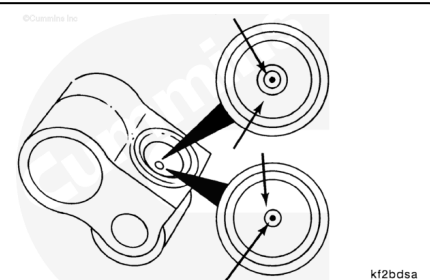
Inspect the cam follower sockets for excessive wear.

If excessive wear is found in the cam follower sockets, the push rods **must** be inspected.

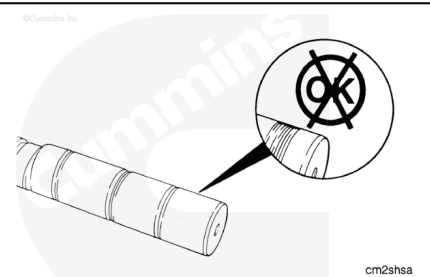
Refer to Procedure 004-014 in Section 4. ([/qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html](http://qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html))

Refer to the Overhead Reuse Guidelines, Bulletin 3810388, to identify wear patterns and excessive wear.

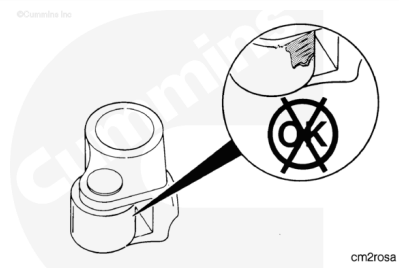
Replace all parts that do **not** meet the reuse guidelines.



Inspect the cam follower shafts for scores or damage.

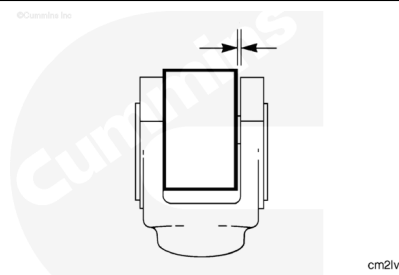


Inspect the cam follower rollers for flat spots, freedom of rotation, or other damage.

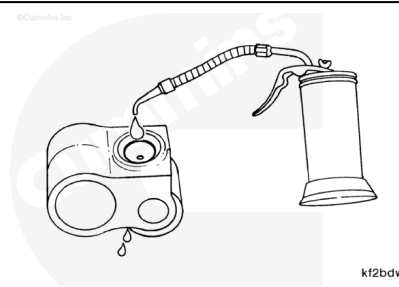


Measure the cam follower roller side clearance.

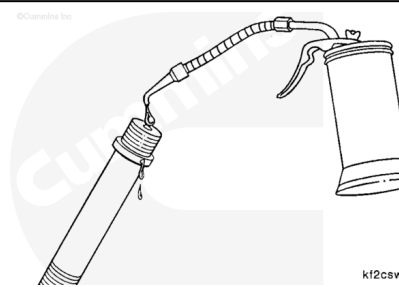
Cam Follower Roller Side Clearance		
mm		in
0.19	MIN	0.007
0.65	MAX	0.026



Use clean 15W-40 engine oil to check the oil flow through the cam followers.

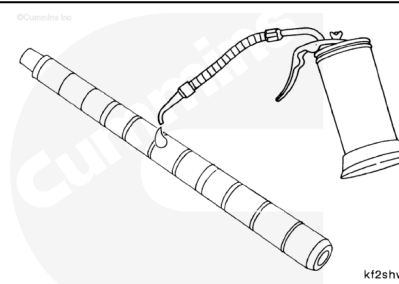


Use clean 15W-40 engine oil to check the oil flow through the cam follower studs.



Assemble

Use clean 15W-40 oil to lubricate the cam followers and shafts.

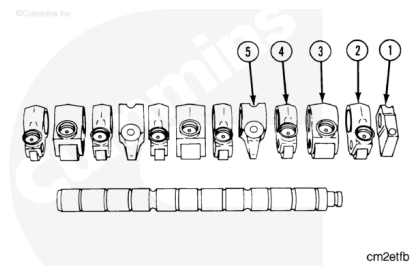


The cam followers, supports, and shafts **must** be installed in the same position from which they were removed. The shaft end supports are **not** interchangeable.

Install the end support (1) on the shaft.

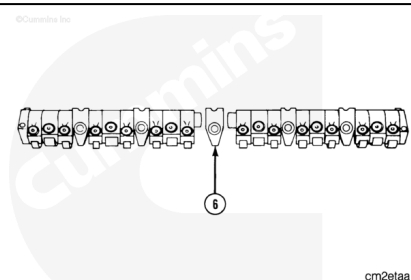
Install the cam followers on the shaft in the following sequence:

- Valve cam follower (2)
- Injector cam follower (3)
- Valve cam follower (4)
- Shaft support (5).



Install the two cam follower shaft assemblies in the common center support (6).

Note : The common center support (6) has a locating dowel hole. The other inner supports do not have this hole.



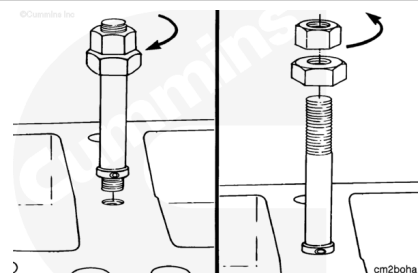
Install

Install the cam follower studs with two M10 x 1.50 nuts tightened together.

Tighten the studs.

Torque Value: 34 n•m [25 ft•lb]

Remove the two nuts.

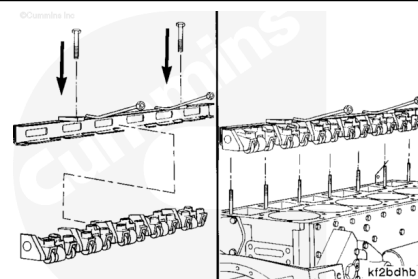


Install cam follower assembly installation and removal tool, Part Number 3824519, onto the cam follower assemblies.

Before installing the cam follower assemblies, the cylinder head deck **must** be cleaned.

Refer to Procedure 002-004 in Section 2.

(/qs3/pubsys2/xml/en/procedures/35/35-002-004-tr.html)



Supports one, four, and seven are doweled. The supports **must** align with the dowel pins in the cylinder block.

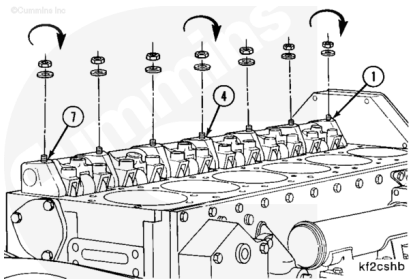
Install the cam follower assemblies on the studs.

Remove the service tool.

Install the washers and nuts on each stud.

Tighten the nuts on supports one, four, and seven.

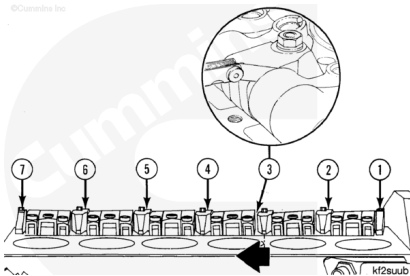
Torque Value: 47 n•m [35 ft-lb]



Install a 0.76-mm [0.030 in] feeler gauge between the number three support and the exhaust lever for cylinder number three. Push the number three support toward the No. four support.

Tighten the number three support nut.

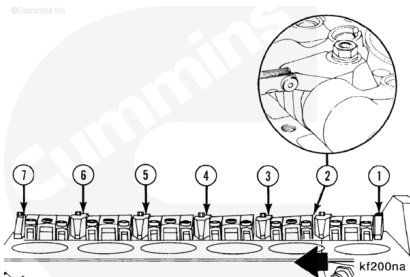
Torque Value: 47 n•m [35 ft-lb]



Install a 0.76-mm [0.030 in] feeler gauge between the number two support and the intake lever for cylinder number two. Push the number two support toward the number three support.

Tighten the number two support nut.

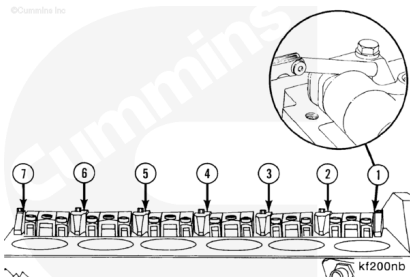
Torque Value: 47 n•m [35 ft-lb]



Measure side clearance between the number one and two supports.

Cam Follower Side Clearance Between Supports

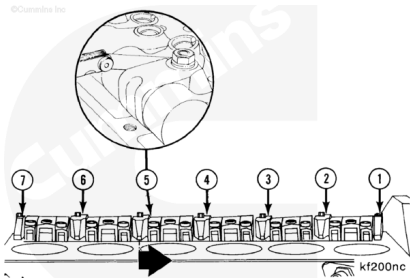
mm		in
0.76	MIN	0.030



Install the 0.76-mm [0.030 in] feeler gauge between the number five support and the exhaust lever for cylinder number four. Push the number five support toward the number four support.

Tighten the number five support nut.

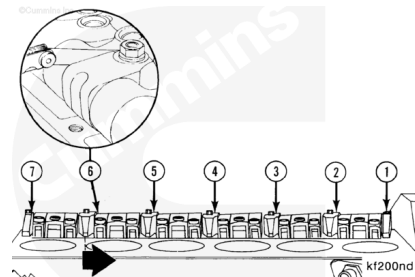
Torque Value: 47 n•m [35 ft-lb]



Install the 0.76-mm [0.030 in] feeler gauge between the number six support and the intake lever for cylinder number five. Push the number six support toward the number five support.

Tighten the number six support nut.

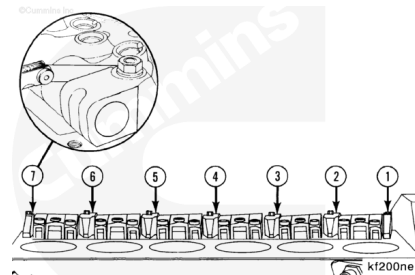
Torque Value: 47 N•m [35 ft-lb]



Measure the side clearance between the number six and seven supports.

Cam Follower Side Clearance Between Supports

mm		in
0.76	MIN	0.030



Finishing Steps

- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/35/35-002-004-tr.html>)
- Operate the engine and check for leaks.

